



DATA STRUCTURE AND ALGORITHM.

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PRESENTATION OF ASSIGNMENT:

SYSTEM NAME: PARCEL DELIVERY SERVICE.

Q62. Parcel delivery service: Stack for undoing delivery orders, queue for processing deliveries, and list to track the parcel.

OPTION CHOICE FOR DELIVERY SERVICE

Output for choose

```
PS C:\Users\user1> & C:/Users/user1/AppData/Local/Programs/Python/Python39-64/Python.exe C:/Users/user1/AppData/Local/Programs/Python/Python39-64/Scripts/Python.exe C:/Users/user1/AppData/Local/Programs/Python/Python39-64/Scripts/Python.exe
No parcels being tracked. 1

PARCEL DELIVERY SERVICE(2024-11-28 15:36:06.604074)

-----
1. Available parcel 2
2. Undo Last Parcel
3. Process Delivery
4. Show Tracked Parcels
5. Exit
-----
Chose from above:
```

Explanation

- For the first time after running codes, there are no parcel being tracked(as for 1) because we need to choose from available parcels(in the stack) before tracking any parcel.
- Second is to see available parcels at option 1, for symbol (2) stored in a stack for tracking that or supplying by any supplier.

SHOWING AVAILABLE LIST OF PARCELS IN THE STACK

Output for viewing the available parcel

```
-----
Chose from above: 1 3
Stored Parcels:
1. Parcel ID: P001, Recipient: Adolphe
2. Parcel ID: P002, Recipient: Evode
3. Parcel ID: P003, Recipient: Bernard
4. Parcel ID: P004, Recipient: Annett
Select a parcel number to add to the delivery queue (1-4):
```

Explanation

- After choosing 1 as it appeared on symbol(3), there are a list of available parcels to be tracked in different place by suppliers.
- Next is to choose from the list(symbol 4), what parcel to be tracked by any supplier to be stored in a queue ready to be tracked.
- Let, as supplier choose parcel at 3

Adding any parcel to the queue to be delivered.

Output after choice of desired parcel by supplier.

```
Select a parcel number to add to the delivery queue (1-4): 3
Parcel P003 added to delivery queue.

Current parcel Your Selected : {'Parcel ID': 'P003', 'Recipient': 'Bernard'}
Do you want to add another? yes/no |
```



Explanation

- You see on symbol(**5**), supplier's choice just added to the queue ready to be tracked or supplied.
- A supplier may like to supply one or more. If he/she wants to add another he/she chooses yes, either choose no as it appears on symbol (**6**) if he/she wants to supply at one parcel.
- Let, as supplier chooses no if he/she wants to supply at one parcel.

RETURNING AN OPTION TO CHOOSE ON.

Output after a supplier type no,

```
Do you want to add another? yes/no no 7
PARCEL DELIVERY SERVICE(2024-11-28 15:36:06.604074)

-----
1. Available parcel
2. Undo Last Parcel 8
3. Process Delivery
4. Show Tracked Parcels
5. Exit
```



Explanation

- After typing no as at symbol(7);
The system return the Options.
But supplier my think that he/she made a mistake when choosing parcel.
The option available is to undo what he/she was just selected.
- Let, he/she make undoing as at symbol(8).

UNDOING THE LAST PARCEL SELECTED.

Output after undoing

```
Last parcel with ID P003 has been undone. 9
```

```
PARCEL DELIVERY SERVICE(2024-11-28 16:43:16.332333)
```

```
-----  
1. Available parcel A  
2. Undo Last Parcel  
3. Process Delivery  
4. Show Tracked Parcels  
5. Exit
```

```
-----  
Chose from above: |
```

Explanation

- After selecting an option 2 to undo, Here the last parcel added had been undone. As at symbol (9).
- Because, we undid selected parcel from our queue, there is no parcel to be delivered. Means, supplier must select again from stored parcels by selecting viewing the available parcels. As at symbol(A).

Showing parcels being tracked after undoing

Output when a user try to show parcel after undoing.

```
-----  
Chose from above: 4  
the tracked parcels are:  
No parcels being tracked.  
  
PARCEL DELIVERY SERVICE(2024-11-29 09:51:11.090403)  
  
-----  
1. Available parcel  
2. Undo Last Parcel  
3. Process Delivery  
4. Show Tracked Parcels  
5. Exit  
  
-----  
Chose from above: █
```



Explanation

- After undoing, there are no parcel to be tracked, because of we added one parcel before and after we undoing it.
- Here we are going to select again any parcel to be delivered and process them/it.

Process delivery of parcel service

Output for processing

```
-----
Chose from above: 3
deque([[...], {'Parcel ID': 'P002', 'Recipient': 'Evode'}, [B]])
PARCEL DELIVERY SERVICE(2024-11-28 16:43:16.332333)

-----
1. Available parcel
2. Undo Last Parcel
3. Process Delivery
4. Show Tracked Parcels
5. Exit

-----
Chose from above: |
```



Explanation

- At symbol(**B**), You see that the Parcel added to the queue ready to be supplied is being processed when you select an option 3(process delivery).
- Next is to show the tracked parcels, being processed from the queue by selecting an option at 4(show tracked parcels).

Showing the tracked parcels

Output after selecting option 4

```
-----  
Chose from above: 4  
the tracked parcels are:  
Tracking the following parcels:  
Parcel ID: P003, Recipient: Bernard (c)  
  
PARCEL DELIVERY SERVICE(2024-11-28 17:28:14.082823)  
  
-----  
1. Available parcel  
2. Undo Last Parcel  
3. Process Delivery  
4. Show Tracked Parcels  
5. Exit  
  
-----  
Chose from above: █
```

Explanation

- After selecting an option 4, the system shows you the parcel which was selected before processing and undoing.
- At symbol (c), that is the parcel being selected to be delivered.
- When the supplier want to exist this system, he/she select the option 5(Exist) to exist the system.

EXISTING THE PARCEL DELIVERY SYSTEM

Output when a user select an option to exist the system

```
-----  
Chose from above: 5  
Exiting the Parcel Delivery Service.  
PARCEL DELIVERY SERVICE(2024-11-28 17:28:14.082823)
```

Explanation

- When a user select to exist, the system stopped.

END... “HAPPY NEW YEAR 2025!”